

Practicing geopandas with Vélib data

Lino Galiana

Version of September 29, 2026

This chapter illustrates the features of `GeoPandas` using the bicycle counts provided by the City of Paris as `open data`. It extends the previous chapter with data that is a bit more complex to handle.

⚠ Automatically generated PDF. This document is produced from the course website and may contain formatting issues. For an up-to-date version adapted to your reading, visit the course page: pythonds.linogaliana.fr.

Table of contents

Informations additionnelles 1

The `GeoPandas` part has recently evolved. You can find the content related to `GeoPandas` in the following chapter (lectures and exercises):

- [Introduction to spatial data with `Geopandas`](#)

Informations additionnelles

i Python environment

This site was built automatically through a `Github` action using the `Quarto` reproducible publishing software (version 1.10.18).

The environment used to obtain the results is reproducible via `uv`. The `pyproject.toml` file used to build this environment is available on the [linogaliana/python-datascientist](#) repository

```
[project]
name = "python-datascientist"
version = "0.1.0"
description = "Source code for Lino Galiana's Python for data science course"
readme = "README.md"
requires-python = "≥3.13,<3.14"
dependencies = [
    "altair≥6.0.0",
    "cartiflette",
    "contextily=1.6.2",
    "duckdb≥0.10.1",
    "folium≥0.19.6",
    "gdal=3.11.4",
    "graphviz=0.20.3",
    "great-tables≥0.12.0",
    "gt-extras≥0.0.8",
    "ipykernel≥6.29.5",
```

```

"jupyter ≥ 1.1.1",
"jupyter-cache ≥ 1.0.0",
"kaleido ≥ 0.2.1",
"langchain-community ≥ 0.3.27",
"loguru = 0.7.3",
"markdown ≥ 3.8",
"nbclient ≥ 0.10.0",
"nbformat ≥ 5.10.4",
"nlTK ≥ 3.9.1",
"pandas ≥ 3.0",
"pip ≥ 25.1.1",
"plotly ≥ 6.1.2",
"plotnine ≥ 0.15",
"polars ≥ 1.8.2",
"pyarrow ≥ 17.0.0",
"pynsee ≥ 0.1.8",
"python-dotenv ≥ 1.0.1",
"python-frontmatter ≥ 1.1.0",
"pywaffle ≥ 1.1.1",
"requests ≥ 2.32.3",
"scikit-image ≥ 0.24.0",
"scikit-learn ≥ 1.8.0",
"scipy ≥ 1.13.0",
"seaborn ≥ 0.13.2",
"selenium < 4.39.0",
"spacy ≥ 3.8.4",
"webdriver-manager ≥ 4.0.2",
"wordcloud = 1.9.3",
]

[tool.uv.sources]
cartiflette = { git = "https://github.com/insee-fr/lab/cartiflette" }
gdal = [
    { index = "gdal-wheels", marker = "sys_platform == 'linux'", },
    { index = "geospatial-wheels", marker = "sys_platform == 'win32'", },
]

[[tool.uv.index]]
name = "geospatial-wheels"
url = "https://nathanjmcDougall.github.io/geospatial-wheels-index/"
explicit = true

[[tool.uv.index]]
name = "gdal-wheels"
url = "https://gitlab.com/api/v4/projects/61637378/packages/pypi/simple"
explicit = true

[dependency-groups]
dev = [
    "nb-clean ≥ 4.0.1",
]

```

To use exactly the same environment (version of `Python` and `packages`), please refer to the [documentation for uv](#).